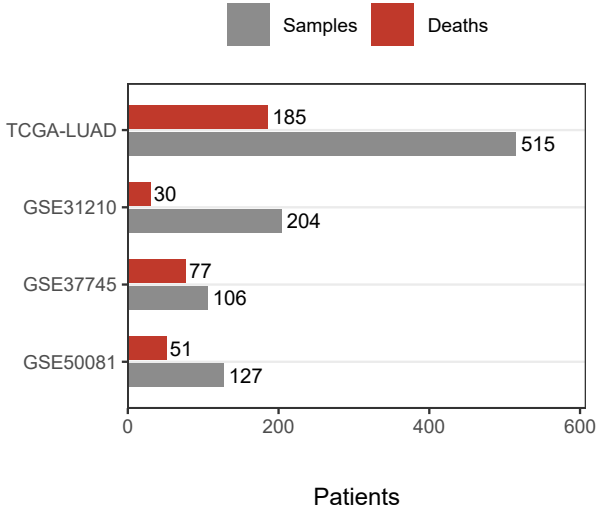
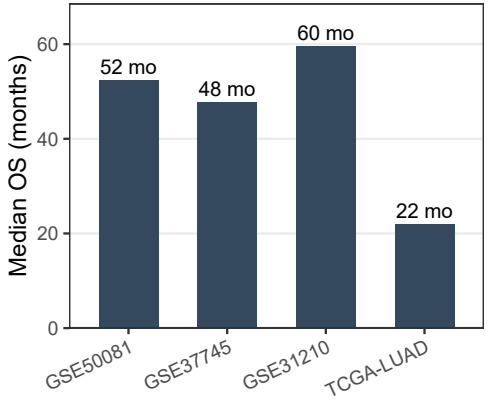
A



B



C

Analysis plan

- Discovery: TCGA-LUAD (n = 515, 185 deaths)
- 4 pre-specified Hallmark programmes -> ssGSEA score per sample
- Validation: 3 independent GEO cohorts (n = 437, 158 deaths)
- Incremental value + negative controls + sensitivity (21 variants)